ARCHIT GUPTA

Apex, NC 27502, 919-432-2583, 919-432-2583, architgupta0602@gmail.com

PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience building Generative AI, agentic automation, machine learning, and backend solutions across quality engineering, cybersecurity, and enterprise applications. Experienced with Python, LLM workflows, REST APIs, Playwright automation, MCP integrations, and AI-powered document processing. Comfortable working across technical and client-facing teams to turn business requirements into practical AI and automation solutions. Interested in building scalable AI systems that improve efficiency, reduce manual work, and solve real-world problems.

SKILLS

- Agentic AI
- JIRA, Smartsheet, SpiraPlan
- Automation Testing
- Python
- Github Copilot, Claude Code, Codex
- Playwright Automation
- Agile
- Client Relations
- Custom MCP

WORK EXPERIENCE

Generative AI Architect — August 2026 - Current
QualityAI

- Leading the development of **QEAgents**, an agentic AI-powered quality engineering and test automation pipeline for Ellucian Banner and Colleague, achieving a **90% reduction in test automation costs** through the transition from traditional automation workflows to AI-driven agent orchestration.
- Architected a multi-agent AI pipeline comprising specialized agents for requirements analysis, test planning, automated test generation, self-healing test execution, and intelligent failure triage.
- Developed custom **Model Context Protocol (MCP) integrations and tooling**, connecting AI agents with Jira, Smartsheet, and SpiraTest through REST APIs to enable intelligent access to requirements, test management systems, and QA artifacts.
- Engineered Playwright-based **agentic automation** workflows to autonomously generate TypeScript test scripts, analyze execution failures, and support automated test healing and pull request generation.
- Designed a local-first agent execution architecture integrating **Claude Code CLI, AWS Bedrock, custom MCP services**, Playwright, GitHub, and Jenkins CI/CD workflows, incorporating human-in-the-loop validation checkpoints.

Data Science and Artificial Intelligence Intern — May 2022 - August 2026
Qualitest Group

- Architected an end-to-end **GenAI document extraction platform** to automate extraction of 16 structured NIAD fields from complex PDF and DOCX syndicate packages and term sheets using schema-guided LLM workflows.
- Designed a **multi-stage LLM architecture** incorporating document routing, section-level extraction, prompt engineering, deterministic validation, and business-rule enforcement to improve extraction accuracy and reliability.
- Implemented a **human-in-the-loop feedback and retrieval workflow** using PostgreSQL/pgvector embeddings, enabling reviewer corrections to be reused as few-shot examples and continuously improve future extractions without model retraining.
- Designed secure **REST API-based processing pipelines** with TLS/OAuth2, document de-duplication, in-memory document processing, structured JSON responses, and a no-document-retention architecture to support secure integration with downstream applications.
- Optimized GenAI inference performance through **parallel LLM execution, prompt caching, selective page routing, and model tiering**, reducing unnecessary token usage, latency, and per-document processing cost while reserving higher-capability models for ambiguous extraction tasks.

- Evaluated and benchmarked **OpenAI and Anthropic LLM models** across accuracy, latency, token consumption, and cost to define a scalable model-selection strategy for production document-processing workloads.

Customer Success Manager Intern December 2025 - May 2026
AccelQ
- Consulted with clients to understand web application functionality, assess testing requirements, and recommend tailored test plans and software quality services.
- Collaborated with client stakeholders and cross-functional teams, **translating technical requirements** into practical testing recommendations and communicating solution value.

Software Engineering Intern May 2025 - July 2025
Anomali Inc. | Redwood City, CA
- Spearheaded a malware attack-flow extraction feature mapping attack patterns, threat actors, and techniques to MITRE ATT&CK identifiers.
- Developed **Python threat-detection algorithms** for a scalable cybersecurity application and contributed backend functionality using Flask, REST APIs, and JavaScript.

Contingent Machine Learning Consultant July 2023 - March 2024
Meta
- Developed Python-based machine learning algorithms for intelligent task routing, applying Natural Language Processing (NLP) and **cosine similarity and re-ranking algorithms** to support task classification and assignment.
- Leveraged big data analytics to optimize task-triaging workflows, achieving a **78% reduction** in triage time and improving operational efficiency.
- Delivered high-priority machine learning projects within Agile methodologies, coordinating execution and priorities across cross-functional teams.
- Partnered with stakeholders to improve task-routing processes through data-driven analysis, algorithm development, and workflow optimization.

EDUCATION

Bachelor Of Science - Computer Science 05/2026
Virginia Tech, Blacksburg, VA, VA

PROJECTS

QEAgents | Agentic AI-Powered QA Automation | Ellucian Banner & Colleague
• Reduced test automation costs by 90% by transitioning QA workflows to an agentic pipeline spanning requirement analysis, test planning, test generation, test healing, and failure triage.
• Built custom local MCP integrations for Jira, Smartsheet, and SpiraTest, alongside Playwright MCP and GitHub tooling, to connect agents with enterprise systems and automation artifacts.
• Implemented specialized AI agents and human-in-the-loop checkpoints, with local execution, AWS Bedrock inference, Git-based artifact handoffs, and Jenkins CI/CD triggers.
DBWorkout | LLM-Powered SQL Learning Platform
• Developed an internal learning platform for Virginia Tech and the University of Virginia, combining an interactive SQL sandbox with an LLM-powered learning coach.
• Designed functionality for instructors to upload PDF materials and generate learning modules with knowledge checks and practical SQL exercises.
QualiHire | AI-Powered Resume & Skills Matching
• Built a resume skill-matching application that scores candidate and employee resumes against employer-submitted job requirements.
• Enabled recruiters and internal management to review candidate fit and support hiring decisions and internal resource alignment.

CERTIFICATIONS

- AccelQ Certification, AccelQ

- Azure Cloud Computing Fundamentals, Microsoft
- Stanford Data Science in Python, Coursera
- Machine Learning and Data Science in Python, Udemy